

Jack mackerel MSE: technical information and work remaining

Scope

This paper allows a reviewer to find the CMP description, OM, performance measure, and results that support the [full MSE report](#) and the [SC14 questions and tasks paper](#). It also distinguishes completed work from scientific and practical work that remains.

Which files control the results

The official set of files is organized in this order:

1. the saved source code, CMP settings, simulation settings, and complete runs;
2. the checked result files, including the 500-draw Slick file;
3. the CMP list, tables, figures, and scorecard inputs made from those results;
4. the explanation in the technical and main papers; and
5. the CMP and annual-use pages made from the same settings.

The current status is documented on the [sources and changes page](#). The final version will include a file list recording the code version, model runs, paper files, software used, file checksums, and products that are no longer current.

Where to find the supporting information

Table 1: Where to find the information needed for each decision.

Topic	Current detailed source	Additional information needed
Development from SCW17	Current report: development	Candidate translation table showing what changed and when
OM set and simulation	Operating models	Complete OM definitions, plausibility rationale, and decision role
Index errors	Generating OM data indices	Calculations, values, data changes, overlapping years, uncertainty, and checks against the observed indices
CMP definitions	CMP list and individual CMP pages	One table with names, roles, units, settings, and file version
Tuning	Candidate procedures	Exact quantity used for tuning, result for every CMP, uncertainty due to the number of runs, and effect of small changes to the trigger
Performance measures	Performance evaluation	Calculations, time periods, how results are combined, whether high or low values are preferred, and how each measure is used
Alternative OMs	Main findings	CMP-by-OM biological performance table, ranges of results, and results for each component in the two-stock OMs
Banking and borrowing	Current analysis	Matched with/without comparison using identical OM draws before implementation conclusions
Annual calculation	Annual workflow and input data	Complete instructions for calculating advice from every shortlisted CMP
Exceptional circumstances	Current framework	Indicators, thresholds, false alarms, missed detections, authority, and response options

Results to report for each CMP and OM

For every CMP and OM or stock-component combination, the final results will report at least:

- mean dynamic Kobe-green probability for the defined tuning period;
- probability below the agreed biomass limit by year, maximum annual risk, and probability of crossing the limit at least once;
- dynamic Kobe-red or other agreed joint-risk statistic;
- overfishing probability;
- near-, medium-, and long-term catch;
- IACC and mean annual reduction in advice;
- probability of catch below 270 kt and any shutdown statistic;
- catch-rate or effort-feasibility measures and frequency of a binding effort cap; and
- whether the CMP is ready for annual use and the reason for any remaining problem.

The results will show uncertainty caused by the number of simulation runs. The paper will explain how uncertainty in the assessment and future stock and index conditions was included, whether CMPs were tested using the same random draws, and whether the conclusions change when more simulations or different random draws are used.

Separate biological requirements from management preferences

The technical paper keeps three steps distinct:

1. **Minimum biological performance:** the requirements that determine whether a CMP can advance.
2. **Alternative assumptions:** performance and plausibility across alternative OMs, including component-specific outcomes.
3. **Management preferences:** trade-offs among candidates that meet the requirements in the first two steps.

The scorecard belongs only in the third step. Its measures, scaling, weights, candidate set, and comparison tests will be saved in downloadable files. It will show how rankings change when OMs, measures, periods, candidates, weights, or random draws change.

Spatial structure and movement

The technical paper reports northern and southern components separately for every two-stock OM and shows the regional indices used by the CMP. It explains:

- the fixed catch splits used to test different assumptions;
- the `split.is` method used in the CMP definitions; and
- any separate northern assessment or management process relevant to interpreting the results.

The allocation work is treated as complete for the present analysis. This section documents what was tested and why; it does not open a new line of allocation analysis. The author's response to the reviewer provides the interpretation to accompany it.

Instructions for annual calculation

Each shortlisted CMP will have one complete set of instructions containing its indicator equation, HCR equation, allocation rule, parameters, data rules, fallback rules, calculation sequence, and ECP interface. A saved retrospective example will include the source data, intermediate values, unrounded advice, annual-limit calculation, allocation, rounding, and final result.

An automatic test will run the example independently and compare every intermediate and final value with saved expected results. Explanatory text may explain the CMP but does not redefine its settings.

Checks before publication

Publication should stop if any of the following disagree:

- CMP membership, names, roles, triggers, targets, minima, or annual limits;
- report, CMP-list, CMP-page, and annual-use-page settings;
- results from the main and alternative OMs;
- tuning probabilities and reference-point definitions;
- two-stock component labels or results;
- scorecard inputs and generated rankings;
- downloadable CSVs, figures, Slick contents, and report statements; or
- file checksums and the final file list.

Work remaining after SC14

Unless completed before submission, the paper records:

- final operational units, cut-offs, missing/revised-data rules, and rounding;
- matched banking-and-borrowing evaluation;
- advice-versus-realized-catch simulation or justification;
- simulation-precision analysis;
- final performance measures and agreed biological requirements;
- tests of whether the ECP detects problems, and rules for who decides what happens next; and
- an independent check that the annual calculation can be repeated.

Each item should have a named person or group responsible for it, a clear test for completion, and a date for decision before a CMP is recommended for adoption.